Paper 20 - Layer-2 Synthesis Ledger

CQE-paper-20
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Construction Basis

Aggregate prediction and theory-admission results into a ledger of solved, open, failed, and transported routes.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-20.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-20\paper_kernel_manifest.json
- body: production\papers\CQE-paper-20\PAPER-BODY.md
- source: production\papers\CQE-paper-20\SOURCE.md
- formal: production\papers\CQE-paper-20\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-20\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-20\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-20.md

Paper 20 - Layer-2 Synthesis Ledger

Abstract

This paper defines the Layer-2 synthesis ledger: the suite-level accounting surface that aggregates solved, open, failed, forbidden, and transported rows without changing their source-paper status. The closed result is ledger behavior. The seed ledger verifies, the rank-24 terminal registry contains twenty-four forms, reachability distinguishes `yes_with_template_glue`, `no`, and `unknown`, the transport-obligation verifier returns `pass_with_open_lifts`, and the contributions registry admits rows only through a named validator.

The paper does not re-prove any source-paper claim. Aggregation is accounting, not automatic closure.

Claims

Claim 20.1. The seeded morphism ledger verifies its internal invariants.

Claim 20.2. The ledger contains a populated object/edge/terminal summary with twenty-four registered terminal forms.

Claim 20.3. Ledger reachability preserves status distinctions: `yes_with_template_glue`, `no`, and `unknown`.

Claim 20.4. The transport layer contains four rows, two demonstrated and two open lifts, with verdict `pass_with_open_lifts`.

Claim 20.5. The contributions registry accepts a durable row only after a named validator accepts it, and records rejected proposals.

Definitions

A **ledger row** is a typed record carrying a source, target, status, classification, witness, and proof boundary.

A **synthesis ledger** is the suite-level collection of those rows.

A **transported row** is useful but not closed; it carries its open boundary with the row.

A **forbidden row** is a retained obstruction, not discarded data.

An **unknown row** is an obligation to seed, refute, or prove a path.

Theorem 20

The Layer-2 synthesis ledger is a verified accounting surface:

```
source receipt -> ledger row -> preserved status -> aggregate report
```

and no aggregate row may be promoted beyond its source evidence.

Proof

The verifier builds a fresh seed ledger and runs `Ledger.verify()`. The result is `status=pass`, proving Claim 20.1.

The ledger summary contains populated object, vector, edge, terminal, discriminant, and obstruction tables. It reports twenty-four terminal forms. This proves Claim 20.2.

The verifier checks three reachability cases. `A1 -> Niemeier:A1^24` returns `yes_with_template_glue`; `G2 -> Niemeier:Leech` returns `no`; an unseeded source returns `unknown`. These are different ledger states and are not collapsed into one verdict. This proves Claim 20.3.

The transport verifier reports four rows, two demonstrated rows, two open lifts, and `all_lifts_demonstrated=false`. This proves Claim 20.4 and keeps open lifts visible.

The registry probe registers a validator that requires a classification field. A valid proposal is accepted and can be looked up; a bare assertion is rejected with a rationale. This proves Claim 20.5.

Together these claims prove the theorem.

Receipt

Promoted verifier:

```
production/formal-papers/CQE-paper-20/verify_layer2_synthesis_ledger.py
```

Receipt:

```
production/formal-papers/CQE-paper-20/layer2_synthesis_ledger_receipt.json
```

Closed layers:

```
seed ledger verifies
ledger summary tables are populated
24 terminal forms are registered
can_close distinguishes yes_with_template_glue, no, and unknown
transport obligations pass with two demonstrated and two open-lift rows
contributions registry accepts validated rows and records rejected rows
```

Open layers:

```
source-paper claims are not re-proved by aggregation
unknown reachability rows remain obligations
forbidden rows remain recorded obstructions
open transport lifts remain open until their source verifiers close them
```

Falsifiers

The paper fails if `Ledger.verify()` fails.

It fails if `unknown` reachability is treated as solved.

It fails if a forbidden row is discarded.

It fails if `pass_with_open_lifts` is promoted to `pass`.

It fails if a contribution enters without validator acceptance.

Role in the Suite

Papers 17, 18, and 19 export verified rows and open residues. Paper 20 is the ledger that keeps those rows visible: solved where solved, open where open, forbidden where forbidden, and transported where transported.

Conclusion

Paper 20 closes the suite's aggregation discipline. It makes the corpus easier to audit by refusing to let aggregation become promotion.